

Phosphorus data

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Reading the data

```
trees<-read.csv("phosphor.csv")
trees
```

```
##      phosphor variety
## 1      0.35    var1
## 2      0.40    var1
## 3      0.58    var1
## 4      0.50    var1
## 5      0.47    var1
## 6      0.65    var2
## 7      0.70    var2
## 8      0.90    var2
## 9      0.84    var2
## 10     0.79    var2
## 11     0.60    var3
## 12     0.80    var3
## 13     0.75    var3
## 14     0.73    var3
## 15     0.66    var3
```

```
is.factor(trees$variety)
```

```
## [1] FALSE
```

We convert the character variable to a factor variable

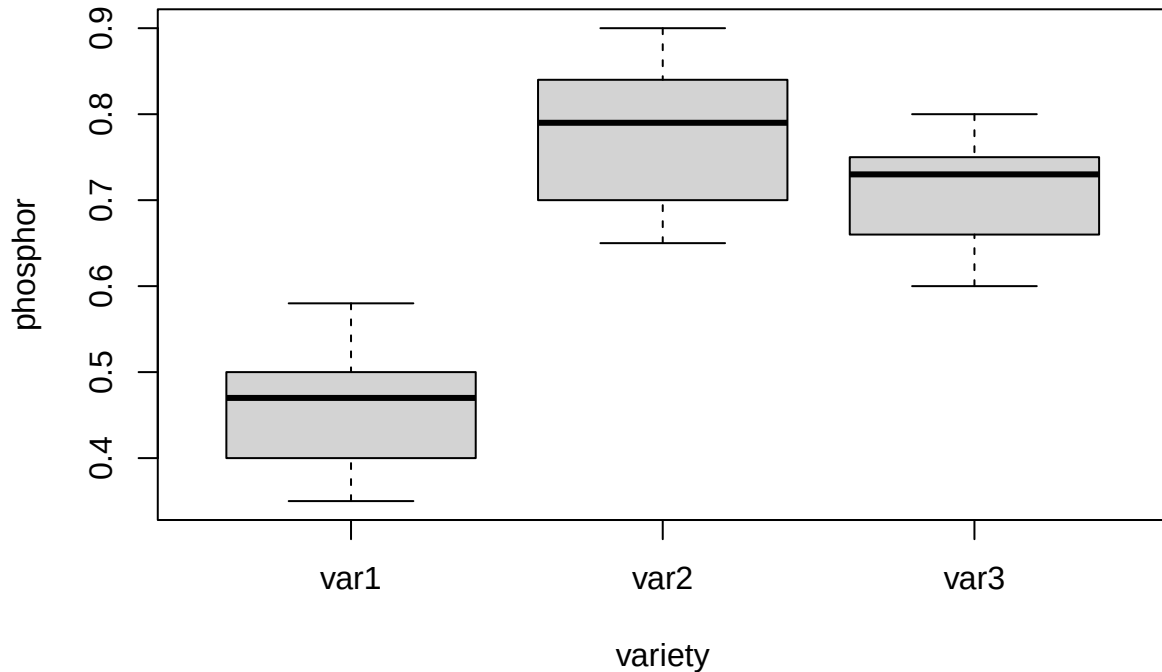
```
trees$variety<-as.factor(trees$variety)
```

check the levels

```
levels(trees$variety)
```

```
## [1] "var1" "var2" "var3"
```

```
plot(phosphor~variety,data=trees)
```



Full model: estimation

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

```
fit.tmp<-lm(phosphor~variety-1,data=trees)
summary(fit.tmp)
```

```
##
## Call:
## lm(formula = phosphor ~ variety - 1, data = trees)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.126 -0.068  0.014  0.053  0.124
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## varietyvar1   0.46000    0.04037   11.39 8.60e-08 ***
## varietyvar2   0.77600    0.04037   19.22 2.22e-10 ***
## varietyvar3   0.70800    0.04037   17.54 6.43e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09028 on 12 degrees of freedom
## Multiple R-squared:  0.9853, Adjusted R-squared:  0.9817
## F-statistic: 268.9 on 3 and 12 DF,  p-value: 2.888e-11
```

Estimated mean levels for each variety

```
coefficients(fit.tmp)
```

```
## varietyvar1 varietyvar2 varietyvar3
##      0.460      0.776      0.708
```

We calculate the mean of each group

```
aggregate(phosphor ~ variety, FUN = mean, data = trees)
```

```
##   variety phosphor
## 1    var1    0.460
## 2    var2    0.776
## 3    var3    0.708
```

Full model: alternative formulation estimation

$$Y_i = \beta_0 + \beta_2 x_{i2} + \beta_3 x_{i3} + \varepsilon_i$$

```
fit<-lm(phosphor~variety,data=trees)
summary(fit)
```

```
##
## Call:
## lm(formula = phosphor ~ variety, data = trees)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.126 -0.068  0.014  0.053  0.124
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.46000    0.04037  11.394 8.6e-08 ***
## varietyvar2  0.31600    0.05710   5.534 0.000129 ***
## varietyvar3  0.24800    0.05710   4.344 0.000956 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09028 on 12 degrees of freedom
## Multiple R-squared:  0.7388, Adjusted R-squared:  0.6953
## F-statistic: 16.97 on 2 and 12 DF,  p-value: 0.0003175
```

Estimated coefficients

```
coefficients(fit)
```

```
## (Intercept) varietyvar2 varietyvar3
##      0.460      0.316      0.248
```

Estimated mean levels for each variety

```
coefficients(fit)+c(0,rep(coefficients(fit)[1],times=2))
```

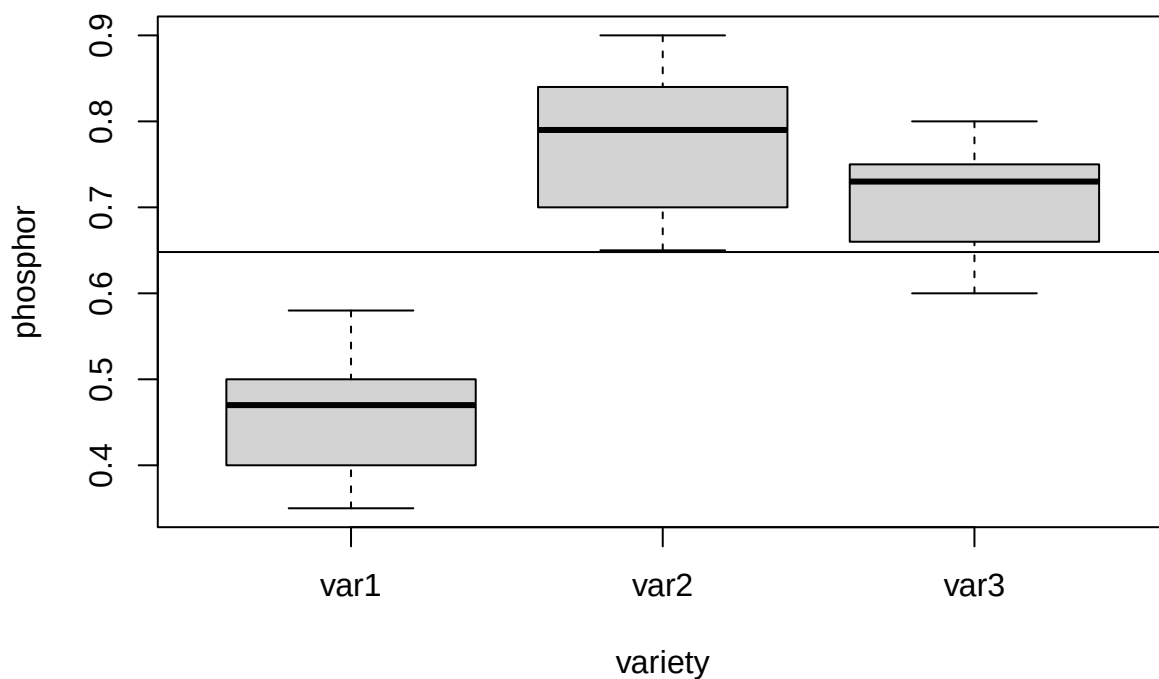
```
## (Intercept) varietyvar2 varietyvar3
##      0.460      0.776      0.708
```

Reduced model estimation

```
fit0<-lm(phosphor~1,data = trees)
summary(fit0)
```

```
##
## Call:
## lm(formula = phosphor ~ 1, data = trees)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.298 -0.108  0.012  0.122  0.252
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.64800    0.04223   15.35 3.76e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1635 on 14 degrees of freedom
```

```
plot(phosphor~variety,data=trees)
abline(h=mean(trees$phosphor))
```



```
mean(trees$phosphor)
```

```
## [1] 0.648
```

Anova F-test

```
anova(fit0,fit)
```

```
## Analysis of Variance Table
```

```
##
## Model 1: phosphor ~ 1
## Model 2: phosphor ~ variety
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      14 0.37444
## 2      12 0.09780  2   0.27664 16.972 0.0003175 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

same result using

```
anova(fit0,fit.tmp)
```

```
## Analysis of Variance Table
##
## Model 1: phosphor ~ 1
## Model 2: phosphor ~ variety - 1
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      14 0.37444
## 2      12 0.09780  2   0.27664 16.972 0.0003175 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```